Problem Set 3

Due Online: Thursday, March 6, 11:59 pm

1. Consider the following Dickey-Fuller regression:

$$\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \varepsilon_t,\ t = 1, 2, \ldots, T.$$

We have learned in class that it is of interest to test $H_0 : \delta = \beta = 0$ against $H_1 : \delta < 0$. Show that if β is not restricted, under the null it will accumulate into a quadratic trend in y_t.

2. Consider the $AR(p)$ model:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t,$$

where $\varepsilon_t \sim IID\left(0, \sigma^2\right)$.

   (a) Represent the model using the lag operator, L, and show that the existence of a unit root $z = 1$ means $\sum_{i=1}^{p} \phi_i = 1$.

   (b) Show that the differenced series $\{\Delta y_t\}$ can be written as

$$\Delta y_t = \phi_1^* y_{t-1} + \phi_2^* \Delta y_{t-1} + \ldots + \phi_p^* \Delta y_{t-p+1} + \varepsilon_t,$$

   where $\phi_1^* = \sum_{i=1}^{p} \phi_i - 1$, and $\phi_j^* = -\sum_{i=j}^{p} \phi_i$, for $j = 2, \ldots, p$. Therefore, testing whether $\{y_t\}$ contains a unit root is equivalent to testing whether $\phi_1^* = 0$.

3. Consider the model for y_t:

$$(1 - L)\, y_t = -1.2 + (1 - 0.6L)\, \varepsilon_t,$$

where $y_0 = 1$ and $\varepsilon_t \sim IID\,(0, 1)$.

   (a) Classify the model as $ARIMA(p, d, q)$.

   (b) Find the mean of y_t, for $t \geq 0$. Does it converge to a finite limit for t large?

   (c) Consider another model: $(1 - 0.95L)\, x_t = -1.2 + (1 - 0.6L)\, \varepsilon_t$, where $x_0 = 1$ and $\varepsilon_t \sim IID\,(0, 1)$. Explain why the mean of x_t converges to a limit for t large and find the limit.

4. Consider the model:

$$y_t = (1 + 0.5L)\left(1 + 0.3L^4\right)\varepsilon_t, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

(a) Is $\{y_t\}$ weakly stationary? Is it invertible?

(b) Find the ACF of $\{y_t\}$.

(c) Compare your finding in part (b) with the ACFs of models $y_t = (1 + 0.5L)\,\varepsilon_t$ and $y_t = \left(1 + 0.3L^4\right)\varepsilon_t$.

5. (Computer Exercise) Let us practice unit root tests on the following series. (All series have been seasonally adjusted.)

(a) Series #1: Logarithm of US quarterly real GDP from 1947Q1 to 2019Q4. You can use the following R code to retrieve the data:

```
library(quantmod)
getSymbols("GDP", src = "FRED")
lgdp < −log(GDP)['/2019']
```

i. Examine the time series plot and ACF of *lgdp*.

ii. Perform the augmented Dickey-Fuller test of the null hypothesis that *lgdp* is difference stationary against the alternative that it is trend stationary.

(b) Series #2: Growth rate of US quarterly real GDP from 1947Q2 to 2019Q4. Use the data in part (a) to generate the GDP growth series as follows:

```
gdpgr < −100 ∗ diff(lgdp)[−1]
```

i. Examine the time series plot and ACF of *gdpgr*.

ii. Perform the augmented Dickey-Fuller test of the null hypothesis of a unit root.

iii. Based on your test results for *lgdp* and *gdpgr*, do you think it is appropriate to apply stationary ARMA models to the difference of log of real GDP?

(c) Series #3: US quarterly inflation rate from 1960Q2 to 2019Q4. First retrieve the quarterly Consumer Price Index (CPI) series from FRED and then compute the inflation rate as the difference of logs.

```
cpi < −getSymbols("CPALTT01USQ661S", src = "FRED", auto.assign = FALSE)
inf < −100 ∗ diff(log(cpi))[−1]
```

i. Examine the time series plot and ACF of inf.

ii. Perform the augmented Dickey-Fuller and KPSS tests to investigate whether inf is stationary.

iii. Perform the augmented Dickey-Fuller and KPSS tests to investigate whether the first difference of inf is stationary.

iv. In practice, many forecasters treat US inflation as $I(1)$. Are your findings consistent with this view?